PAVS: Bridging AMD's Softpower Gap

From the Voice AI Perspective

Why a competitive silicon doesn't win by itself?

Voice AI is the lens -- conclusions Hold

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The Voice AI: Business, Opportunities & Tech-Stack

A Massive Inference Market, Being Decided Now — Next is Edge Inference

The opportunity

Metric	Value
AI voice-agents market, 2034	**$47.5B** @ **35% CAGR**
Cost per call — human vs AI	**$7–12 → ~$0.40**
YoY growth, production deploys	**340%** (500+ orgs)
Contact-center labor savings, 2026	**$80B** (Gartner)
Orgs planning voice AI by end-2026	**80%**

Fig 1 · The five-layer voice pipeline — 3 layers are GPU-compute-bearing *(median E2E 680 ms → sub-500 ms bar)*

1. Telephony / Transport (<50 ms)

▼

2. ASR / STT (<150 ms) · *compute*

▼

3. LLM / NLU (<200 ms) · *compute*

▼

4. TTS (<150 ms) · *compute*

▼

5. Orchestration · VAD · routing · RAG

Shaded **ASR · LLM · TTS** consume GPU cycles — AMD's advantage *and* its gaps both live here.

> **A massive, fast-growing inference market whose compute default is being chosen right now.**

It's a Compute Market at Its Core

A deployed agent is a steady-state inference machine

- It doesn't train — it **transcribes, reasons & speaks 24×7**
- At scale, its **dominant operating cost is GPU compute**
- **3 of the 5** pipeline layers — **ASR · LLM · TTS** — are GPU-compute-bearing

Where the money actually goes

- **LLM** is cheapest per minute but **most latency-critical** — first token in **<200 ms**
- **TTS is the hidden budget killer** — premium hosted synthesis is **$0.10–$0.30/min**, often exceeding ASR + LLM combined → self-hosting matters
- The next frontier is **on-device / edge inference** — kiosks, vehicles, factory floors, hospitals and public services

> **Three of five layers burn GPU cycles every second of every call**

AMD Already Has the Winning Silicon

Table 2 · LLM inference — MI300X vs H100 SXM5 *(Llama 3.1 70B, 2026)*

Dimension		H100 SXM5
VRAM	**192 GB** HBM3	80 GB HBM3e
On-demand cloud price	**$1.50–2.50/hr**	$2.90/hr
Throughput (tok/s)	~18,000	~19,500
Cost / 1M tokens	**$0.027**	$0.041
70B fits on 1 GPU?	**Yes**	No (needs 2)

MI355X (288 GB HBM3e) lands within single-digit % of B200 on MLPerf Inference v6.0.

The savings compound at scale

- **34% lower cost-per-token** → **~$84K/month** saved at 1M min/month
- Self-hosted ROCm TTS removes **$100K–$300K/month** in hosted synthesis
- India on-prem: **~50% lower GPU CapEx**
- Single-GPU fit halves node count — no NVLink / interconnect complexity

The advantage is real — yet unrealized in the Market — This is not a hardware problem.

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The Voice AI Stack Has Converged — One Compute Decision Rules the Market

*Despite hundreds of companies and billions in funding, production code sits above just **three open-source frameworks** — sharing one model layer and one compute decision.*

EVERY VOICE AI COMPANY

Vapi · Retell · Bolna · Gnani · PolyAI · Synthflow · Ringg.ai

LiveKit	Pipecat	TEN
Apache 2.0	BSD-2	C++/Go

SHARED MODEL LAYER

STT: Whisper / Sarvam · LLM: Llama / GPT-4o · TTS: Kokoro / ElevenLabs

GPU COMPUTE NVIDIA / **AMD**

Owning the compute isn't enough — you win by owning the on-ramps just above it: HW choice is made before you reach the compute layer.

How NVIDIA Became the Gateway — The NIM Playbook

*Dominance wasn't silicon alone. The instrument was **NIM** — pre-packaged containers that make NVIDIA the path of least resistance at the first developer touchpoint.*

Ready-to-run microservices

- **Riva ASR NIM** — streaming ASR (Parakeet) in one `docker pull`; no config, no VAD tuning
- **Maxine Studio Voice NIM** — real-time noise suppression / echo cancellation → directly lowers ASR word-error-rate

Complete applications

- **AI Blueprints** — deployable voice-agent repos, not docs. The GPU choice is already encoded inside.

Developer & cloud on-ramps

- **RTX Spark** — Grace+Blackwell devkit, 128 GB shared memory; prototype on NVIDIA → default to NVIDIA in production
- **Azure AI Foundry NIM** (July 2026) — deploy from VS Code in one click, NVIDIA GPU underneath
- **SI certification** — TCS, Infosys, Wipro, Accenture, Capgemini trained & certified on NIM

> **You never consciously select a GPU — Choice is already made. - NVIDIA is the default; every other choice requires active effort. That is softpower.**

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CUDA Gap Closed; AMD Still Lacks a Low-Resistance Path to Its Silicon

Table 1 · ROCm support for voice-AI components (2026)

Component	ROCm	Note
LLM serving (vLLM, Llama 3.1)	Full	90–95% of H100 tput
LLM serving (SGLang)	Full	Within 5–10% of CUDA
TTS: Kokoro, Chatterbox	Full	Pure PyTorch, 0 changes
TTS: Coqui XTTS-v2	Partial	Works with effort
ASR: Faster-Whisper (batch)	Full	RT mode needs work
ASR: streaming containers	**Gap**	CUDA-assumed
Embeddings: HF TEI	**Gap**	No ROCm build

Two heaviest layers (LLM, TTS) solved; streaming ASR + TEI remain open.

Table 3 · NIM ecosystem gap: NVIDIA vs AMD required

NVIDIA Has	AMD Equivalent	Status
Riva ASR NIM	AMD ASR container	**Absent**
Riva TTS NIM	AMD TTS container	**Absent**
AI Blueprint: voice agent	Pipecat on MI300X	**Absent**
Pipecat NVIDIA processor	Pipecat AMD processor	**Absent**
LiveKit certified backend	LiveKit AMD backend	**Absent**
VS Code NIM extension	ROCm / Quark ext.	**Absent**
Azure Foundry NIM	Azure AMD endpoint	Partial
SI cert (TCS, Infosys)	SI cert program	**Absent**
28M devs default CUDA	AMD dev default	Opt-in

Table 4 · Framework ecosystem presence

Framework	NVIDIA	AMD
Pipecat	Official processor	Absent
LiveKit Agents	Certified backend	Absent
LangChain	`langchain-nvidia-ai-endpoints`	Absent
LlamaIndex	`llama-index-llms-nvidia`	Absent
Haystack	NIM pipeline component	Absent

The gap is not silicon — it's the least-resistant path to silicon: containers, plugins, blueprints & marketplace slots.

The Five Downstream Ecosystem Layers

Same logic at every layer: ship a named package → developers adopt the default → the hardware choice becomes invisible at the call site.

1 · Framework plugin slots

`ChatNVIDIA(...)` routes to NVIDIA at the endpoint. AMD doesn't exist at that call site — repeated across millions of pipelines.

2 · Inference serving infrastructure

Triton / Dynamo + TensorRT-LLM = the invisible substrate (20–30% edge at batch 1–4). AMD has **vLLM ROCm only** — one tool, not infrastructure.

3 · Voice-domain SDKs

Riva (ASR+TTS+MT+diarization) · Maxine (noise suppression) · ACE (digital humans). **AMD has no equivalent for any of the three.**

4 · Edge hardware platform

Jetson (Nano → Thor) + JetPack ships Riva, vLLM, CUDA. Certified OEMs: Siemens, KUKA, ADLINK. Caterpillar runs Jetson Thor + Riva inside a bulldozer cab. AMD Ryzen AI = **laptop only** — no industrial module.

5 · Cloud marketplace slots

NIM native on Azure Foundry; selectable on HF, SageMaker, Vertex. AMD **absent / limited** — TGI ROCm image is production-ready but **not surfaced in the UI**.

> **The pattern: ship the default package, and the hardware choice disappears at the call site. AMD is present at none of them.**

The Four Upstream Gaps

Gap 1 · Training & fine-tuning moat

Voice companies fine-tune weekly (custom ASR/TTS/NLU). **NeMo → Triton export** locks the whole loop to NVIDIA.

Gap 2 · ISV & SI sales channel

GPU chosen implicitly by the ISV. NVIDIA is in **Genesys** (11k+ customers), **Cisco Webex** (Maxine), **Salesforce**, **ServiceNow**. SIs (TCS, Infosys, Wipro, Accenture) are **certified on NVIDIA**. AMD: no presence in any major contact-center ISV.

Gap 3 · Voice + vision convergence

Next-gen agents fuse camera + mic. Jetson Thor unifies **Isaac · Metropolis · Riva · ACE** on one substrate. AMD's lines don't converge (MI300X · Ryzen AI · Radeon) — a **missing product category**.

Gap 4 · Synthetic data for low-resource languages

No AMD-native speech-data pipeline or academic partnership (e.g. AI4Bharat) → lifecycle dependency predates any inference decision.

> **Closing downstream puts AMD in contention; closing upstream creates platform momentum.**

The AMD Gateway Strategy

Run NVIDIA's playbook — targeting the cost wedge its pricing creates. Same Pipecat code, same LiveKit transport, 34% lower per-conversation LLM cost.

Step 1 · Three certified containers

`docker pull` → validated in 30 s: (a) Pipecat Voice Agent (Whisper + vLLM + Kokoro), (b) LiveKit + SIP, (c) Sarvam India (DPDP-compliant, on-prem).

Step 2 · AMD Voice AI Blueprints

Deployable repos, not docs — US/EU contact center, India BFSI, Arabic GCC, edge/offline. Must run on a **~$700 consumer GPU**.

Step 3 · Pipecat ROCm plugin

`pipecat-ai[amd]` on PyPI — AMD as an official processor alongside Deepgram / ElevenLabs. **~2 weeks.**

Step 4 · HF TEI on ROCm

Upstream ROCm into Text-Embeddings-Inference — closes the last RAG software gap.

Step 5 · HF Inference Endpoints slot

TGI ROCm is ready — one **business agreement** makes MI300X selectable. **Zero eng weeks.**

Step 6 · Ryzen AI NPU on-ramp

`pipecat create --backend amd-npu` — offline, no GPU, no API key. The dev-laptop demo that mirrors **RTX Spark**.

> **PAVS sits exactly at this intersection — between AIG model outputs and lighthouse customers. It does Implicitly what NVIDIA's platform teams do intentionally**

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Prioritized Action Roadmap

Tier 1 — Pure engineering *(no new silicon, no product decisions)*

Action	Effort	Horizon	Impact
HF Inference Endpoints slot	Business neg.	0–4 wks	High
Certified ASR + TTS containers	2 wks each	1 month	High
Pipecat AMD processor (PyPI)	2 wks	1 month	Medium
LangChain / LlamaIndex / Haystack	2–4 wks ea.	1–2 mo	High
AMD Voice AI Blueprints (4 targets)	4 wks each	2 months	High
`bitsandbytes` ROCm official	4–6 wks	2 months	High
Triton ROCm certified image	12–16 wks	4 months	High
Ryzen AI NPU voice template	4 wks	2 months	Medium

Tier 2 — Business dev + engineering

Action	Effort	Horizon	Impact
Genesys ISV integration	Business dev	6–12 mo	Very high
SI cert program (TCS, Infosys)	Program build	6–12 mo	Very high
FlashAttention 3 / CDNA3 equiv.	16–24 wks	6 months	Medium

Tier 3 — Strategic HW commitments

Action	Effort	Horizon	Impact
AMD voice data initiative (India)	$5–13M, 2 yr	18–24 mo	Strategic
Jetson-equiv. industrial edge module	HW roadmap	2–3 yrs	Very high

First 12–16 weeks: Steps 1–4 with 6–8 engineers make AMD visible at first contact. India BFSI under DPDP is the natural beachhead — on-prem mandatory, ~50% lower CapEx.

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How the Bridge Is Being Closed

The gap can't be closed by a new chip

- It's a **software & ecosystem** problem — containers, plugins, blueprints, marketplace slots
- Which is **exactly** the problem a **vertical-software team** solves

PAVS produces the closing assets organically

- Each lighthouse deployment → a **reusable pipeline** = a step toward an AMD-native ecosystem
- Each model the Physical AI SDK supports → **one less** requiring NVIDIA tooling on AMD HW
- Each robotics pipeline on Ryzen AI APU → a **reference design** for an OEM / SI / ISV

AMD AI Group (AIG)

↓ models + research ↑ deployment feedback

PAVS

↓ ↑

Physical AI SDK	**Multi-modal pipelines**
inference substrate	+ Jetson-class devkit

> **Make AMD the path of least resistance at the first touchpoint, and the rest of the stack follows.**